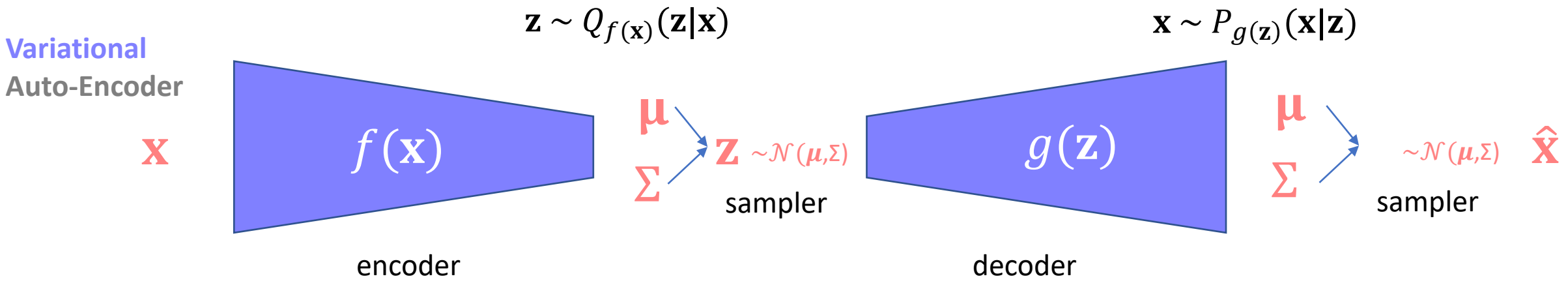
Let’s make it probabilistic and truly generative!

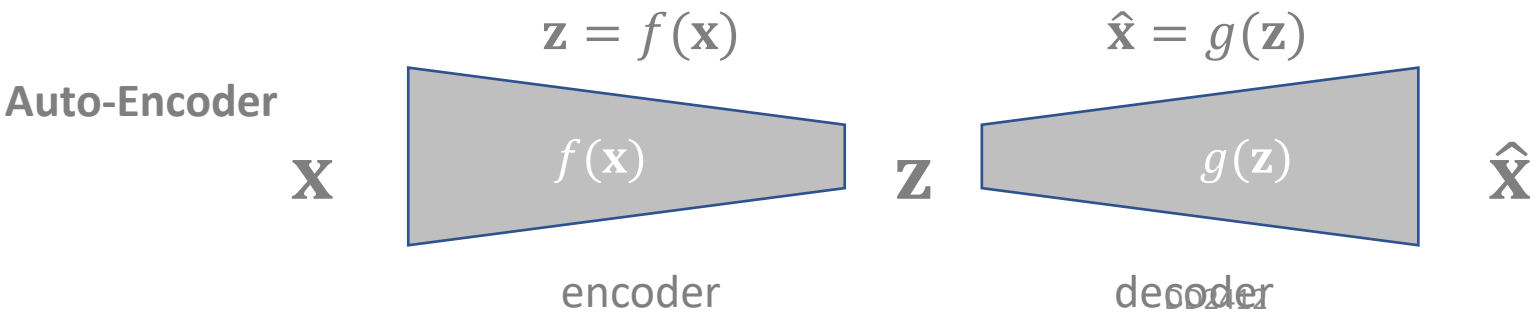
- there are several steps!

Kingma, Welling "**Auto-Encoding Variational Bayes“,** ICLR 2013

Variational AutoEncoder

VAE: implementation perspective



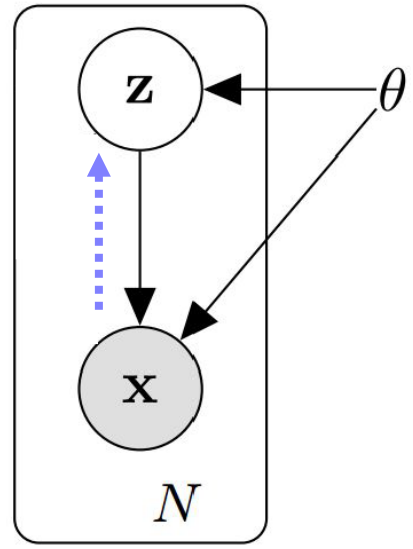


Variational AutoEncoder - Steps

Step 1: Latent Coding

- First: what are we looking for in latent variable generative models?
  - For the generation process:
    - we need to assume a $P(\mathbf{z})$
    - we need to devise a parametrized likelihood $P(\mathbf{x}|\mathbf{z})$
    - Learn the parameters of the likelihood $P(\mathbf{x}|\mathbf{z})$
    - then we can sample $\hat{\mathbf{z}} \sim P(\mathbf{z})$ and then sample $\hat{\mathbf{x}} \sim P(\mathbf{x}|\mathbf{z} = \hat{\mathbf{z}})$
  - For the coding process (if any):
    - we need to infer $P(\mathbf{z}|\mathbf{x})$ using the above pdfs

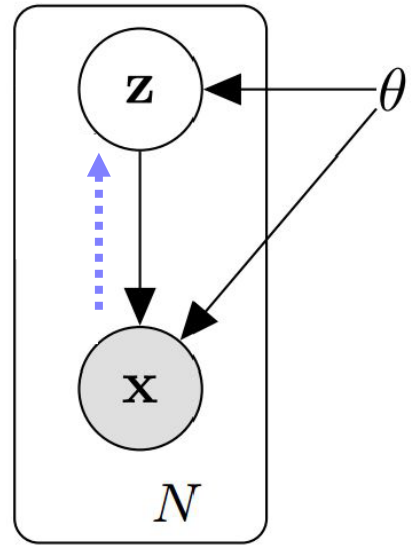
now let's focus on the coding part!



VAE - Steps

Step 1: Latent Coding

- First: what are we looking for in latent variable generative models?
  - For the generation process:
    - we need to assume a $P(\mathbf{z})$
    - we need to devise a likelihood $P(\mathbf{x}|\mathbf{z})$
    - Learn the parameters of the likelihood $P(\mathbf{x}|\mathbf{z})$
    - then we can sample $\hat{\mathbf{z}} \sim P(\mathbf{z})$ and then sample $\hat{\mathbf{x}} \sim P(\mathbf{x}|\mathbf{z} = \hat{\mathbf{z}})$
  - For the coding process (if any):
    - we need to infer $P(\mathbf{z}|\mathbf{x})$ using the above pdfs
    - $P(\mathbf{z}|\mathbf{x}) = P(\mathbf{x}|\mathbf{z})P(\mathbf{z})/P(\mathbf{x})$

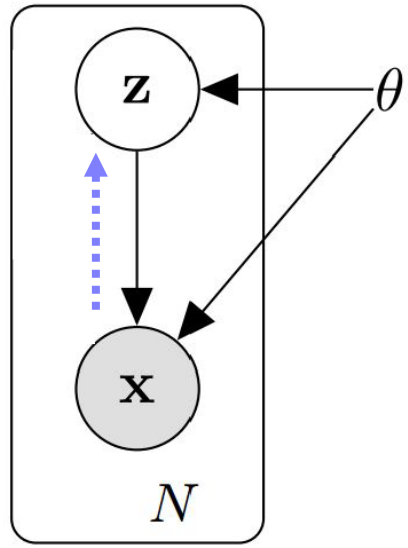


Remember Bayesian inference, intractability, and variational inference

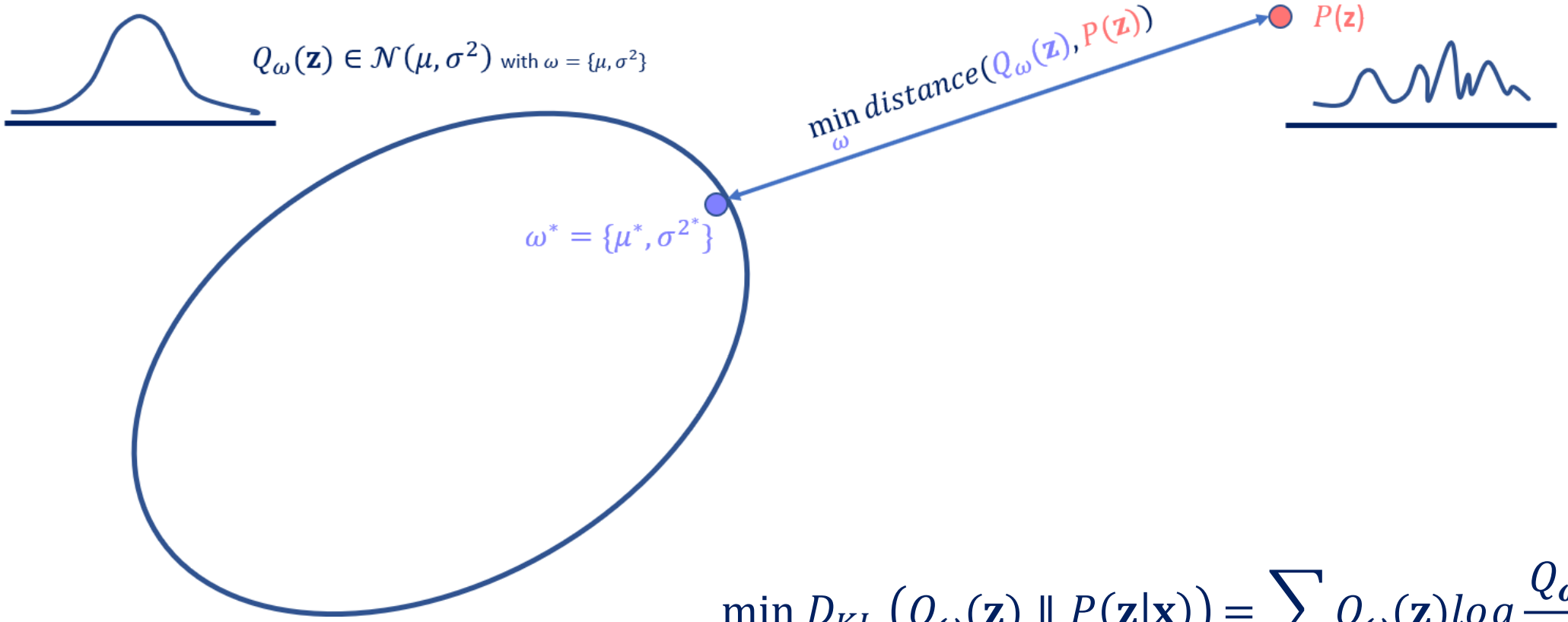
VAE - Steps

Step 1: Latent Coding

- First: what are we looking for in latent variable generative models?
  - For the generation process:
    - we need to assume a $P(\mathbf{z})$
    - we need to devise a likelihood $P(\mathbf{x}|\mathbf{z})$
    - then we can sample $\hat{\mathbf{z}} \sim P(\mathbf{z})$ and then sample $\hat{\mathbf{x}} \sim P(\mathbf{x}|\mathbf{z} = \hat{\mathbf{z}})$
  - For the coding process (if any):
    - we need to infer $P(\mathbf{z}|\mathbf{x})$ using the above pdfs
    - $P(\mathbf{z}|\mathbf{x}) = P(\mathbf{x}|\mathbf{z})P(\mathbf{z})/P(\mathbf{x})$
    - $P(\mathbf{x})$ is interactable
    - VI: we approximate it with $Q_\omega(\mathbf{z}|\mathbf{x})$



VAE - Steps



$$\min_{\boldsymbol{\omega}} D_{KL}\left(Q_{\boldsymbol{\omega}}(\mathbf{z}) \parallel P(\mathbf{z}|\mathbf{x})\right) = \sum Q_{\boldsymbol{\omega}}(\mathbf{z}) log \frac{Q_{\boldsymbol{\omega}}(\mathbf{z})}{P(\mathbf{z}|\mathbf{x})}$$

Step 1: Latent Coding

So, using VI, to find the coding (posterior of latent variables **z**) we do the following optimization:

$$\max_{\boldsymbol{\omega}=\{\boldsymbol{\omega}_i\}_{i=1:n}} \sum_{i=1}^{n} \left(\mathbb{E}_{Q_{\boldsymbol{\omega}_i}(\mathbf{z}|\mathbf{x}_i)} \log P(\mathbf{x}_i|\mathbf{z}) - D_{KL}(Q_{\boldsymbol{\omega}_i}(\mathbf{z}|\mathbf{x}_i)||P(\mathbf{z})) \right)$$

expected likelihood **prior objective**

Reflection/Discussion point

- Remember the VI objective for discriminative deep networks in the last session and compare with the VI objective for inferring the latent codes of today's deep generative models.
- Discuss what is the true distribution and what is the approximating distribution in each case.
- Any other differences?



5 minutes!

$$\max_{\boldsymbol{\omega}} \mathcal{L}_{VI-BNN} = \mathbb{E}_{Q_{\boldsymbol{\omega}}(\boldsymbol{\theta})} \log P(\mathcal{D}|\boldsymbol{\theta}) - D_{KL}\big(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) \parallel P(\boldsymbol{\theta})\big)$$

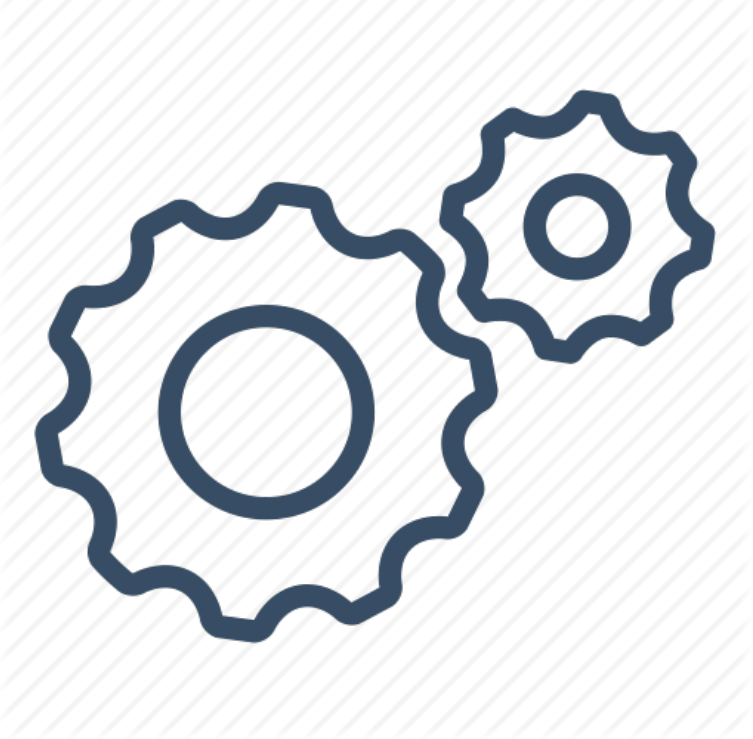
$$\max_{\boldsymbol{\omega}=\{\boldsymbol{\omega}_i\}_{i=1:n}} \mathcal{L}_{VAE} = \sum_{i=1}^{n} \Big(\mathbb{E}_{Q_{\boldsymbol{\omega}_i}(\mathbf{z}|\mathbf{x}_i)} \log P(\mathbf{x}_i|\mathbf{z}) - D_{KL}(Q_{\boldsymbol{\omega}_i}(\mathbf{z}|\mathbf{x}_i)||P(\mathbf{z})) \Big)$$

VAE - Steps

Step 1: Latent Coding

$$\max_{\boldsymbol{\omega}=\{\boldsymbol{\omega}_i\}_{i=1:n}} \mathcal{L}_{VAE} = \sum_{i=1}^{n} \left(\mathbb{E}_{Q_{\boldsymbol{\omega}_i}(\mathbf{z}|\mathbf{x}_i)} \log P(\mathbf{x}_i|\mathbf{z}) - D_{KL}(Q_{\boldsymbol{\omega}_i}(\mathbf{z}|\mathbf{x}_i)||P(\mathbf{z})) \right)$$

$$\max_{\boldsymbol{\omega}} \mathcal{L}_{VI-BNN} = \mathbb{E}_{Q_{\boldsymbol{\omega}}(\boldsymbol{\theta})} \log P(\mathcal{D}|\boldsymbol{\theta}) - D_{KL}\left(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}||\boldsymbol{D}) \parallel P(\boldsymbol{\theta})\right)$$



There are many (n) variational optimizations to be done for the VAE objective!

Step 2: Amortized Variational Inference

- Second: let's get Deep Networks involved

$$\max_{\boldsymbol{\omega}=\{\boldsymbol{\omega}_i\}_{i=1:n}} \sum_{i=1}^{n} \left(\mathbb{E}_{Q_{\boldsymbol{\omega}_i}(\mathbf{z}|\mathbf{x}_i)} \log P(\mathbf{x}_i|\mathbf{z}) - D_{KL}(Q_{\boldsymbol{\omega}_i}(\mathbf{z}|\mathbf{x}_i)||P(\mathbf{z})) \right)$$

  - Assume the variational parameters, $\boldsymbol{\omega}$ of $Q_{\boldsymbol{\omega}}(\mathbf{z}|\mathbf{x})$ can be modelled by an NN ($f(\mathbf{x})$) with parameters θ_f, i.e., $\boldsymbol{\omega}_i = f_\theta(\mathbf{x}_i)$
  - Then we can, instead of directly maximizing for $\boldsymbol{\omega}$, maximize for θ_f:

  - $\max_{\theta_f} \sum_{i=1}^{n} \left(\mathbb{E}_{Q_{f(\mathbf{x}_i)}(\mathbf{z}|\mathbf{x}_i)} \log P(\mathbf{x}_i|\mathbf{z}) - D_{KL}(Q_{f(\mathbf{x}_i)}(\mathbf{z}|\mathbf{x}_i)||P(\mathbf{z})) \right)$

This deep network (f_θ) is the probabilistic encoder!

Step 2: Amortized Variational Inference

- Optimization/Inference 1
  - input 1: $\mathbf{x}_1$
  - parameters 1: $\boldsymbol{\omega}_1$
  - objective 1: $\max_{\boldsymbol{\omega}_1} \mathbb{E}_{Q_{\boldsymbol{\omega}_1}(\mathbf{z}|\mathbf{x}_1)} \log P(\mathbf{x}_1|\mathbf{z}) - D_{KL}(Q_{\boldsymbol{\omega}_1}(\mathbf{z}|\mathbf{x}_1)||P(\mathbf{z}))$
- Optimization/Inference 2
  - input 2: $\mathbf{x}_2$
  - parameters 2: $\boldsymbol{\omega}_2$
  - objective 2: $\max_{\boldsymbol{\omega}_2} \mathbb{E}_{Q_{\boldsymbol{\omega}_2}(\mathbf{z}|\mathbf{x}_2)} \log P(\mathbf{x}_2|\mathbf{z}) - D_{KL}(Q_{\boldsymbol{\omega}_2}(\mathbf{z}|\mathbf{x}_2)||P(\mathbf{z}))$
- …
- Optimization/Inference n
  - input 1: $\mathbf{x}_n$
  - parameters 1: $\boldsymbol{\omega}_n$
  - objective 1: $\max_{\boldsymbol{\omega}_n} \mathbb{E}_{Q_{\boldsymbol{\omega}_n}(\mathbf{z}|\mathbf{x}_n)} \log P(\mathbf{x}_n|\mathbf{z}) - D_{KL}(Q_{\boldsymbol{\omega}_n}(\mathbf{z}|\mathbf{x}_n)||P(\mathbf{z}))$

Instead, learn a function that goes from the input space to the parameters space!

$$f_\theta \colon \mathcal{X} \mapsto \Omega$$

Learn the parameters of that function!

VAE - Steps

Step 3: Likelihood Function

- Third: let's get deep networks involved $\max_{\theta_f} \sum_{i=1}^{n} \left(\mathbb{E}_{Q_{f(\mathbf{x})}(\mathbf{z}|\mathbf{x})} \log P(\mathbf{x}|\mathbf{z}) - D_{KL}(Q_{f(\mathbf{x})}(\mathbf{z}|\mathbf{x}) || P(\mathbf{z}$

  - The likelihood function (generator) is also generally unknown
  - Assume $P(\mathbf{x}|\mathbf{z})$ is a parametrized distribution too, and its parameters are obtained by another NN ($g(\mathbf{z})$) with parameters θ_g
  - Let us try to find its parameters θ_g in the same optimization using all data points:

  - $\max_{\theta_f, \theta_g} \sum_{i=1}^{n} \left(\mathbb{E}_{Q_{f(\mathbf{x}_i)}(\mathbf{z}|\mathbf{x}_i)} \log P_{g(\mathbf{z})}(\mathbf{x_i}|\mathbf{z}) - D_{KL}(Q_{f(\mathbf{x}_i)}(\mathbf{z}|\mathbf{x}_i) || P(\mathbf{z})) \right)$

This Deep Network is the probabilistic decoder

and we are doing an ML estimation of it!

Reflection/Discussion point



- But wait, didn't we start with Variational Inference (approximation of Bayesian modelling)?
- Why do I say the optimization is a maximum likelihood estimation (ML) of the *generative model* ($P(\mathbf{x}|\mathbf{z})$)?

5 minutes!

$$\max_{\theta_f,\theta_g} \sum_{i=1}^{n} \left(\mathbb{E}_{Q_{f(\mathbf{x}_i)}(\mathbf{z}|\mathbf{x}_i)} \log P_{g(\mathbf{z})}(\mathbf{x_i}|\mathbf{z}) - D_{KL}(Q_{f(\mathbf{x}_i)}(\mathbf{z}|\mathbf{x}_i)||P(\mathbf{z})) \right)$$

VAE - Steps

Step 4: Approximating Distribution

$$\max_{\theta_f,\theta_g} \sum_{i=1}^{n} \left(\mathbb{E}_{Q_{f(\mathbf{x})}(\mathbf{z}|\mathbf{x}_i)} \log P_{g(\mathbf{z})}(\mathbf{x}_i|\mathbf{z}) - D_{KL}(Q_{f(\mathbf{x})}(\mathbf{z}|\mathbf{x}_i)||P(\mathbf{z})) \right)$$

- Fourth: what family of approximating distributions are we considering?

  - Prior: $P(\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu} = \mathbf{0}, \Sigma = \mathbb{I})$
  - Likelihood: $P_{g(\mathbf{z})}(\mathbf{x}|\mathbf{z}) = \mathcal{N}\left(\{\boldsymbol{\mu}_g, \Sigma_g\} = g(\mathbf{z})\right)|_{\mathbf{x}}$
  - Posterior: $Q_{f(\mathbf{x})}(\mathbf{z}|\mathbf{x}) = \mathcal{N}\left(\{\boldsymbol{\mu}_f, \Sigma_f\} = f(\mathbf{x})\right)|_{\mathbf{z}}$

  - $\max_{\theta_f,\theta_g} \sum_{i=1}^{n} \left(\mathbb{E}_{\mathcal{N}(\mathbf{z};\{\boldsymbol{\mu},\Sigma\}=f(\mathbf{x}_i))} \log \mathcal{N}(\mathbf{x}_i; \{\boldsymbol{\mu}, \Sigma\} = g(\mathbf{z})) - D_{KL}(\mathcal{N}(\{\boldsymbol{\mu}, \Sigma\} = f(\mathbf{x}_i)) \,||\, \mathcal{N}(\boldsymbol{\mu} = \mathbf{0}, \Sigma = \mathbb{I})) \right)$

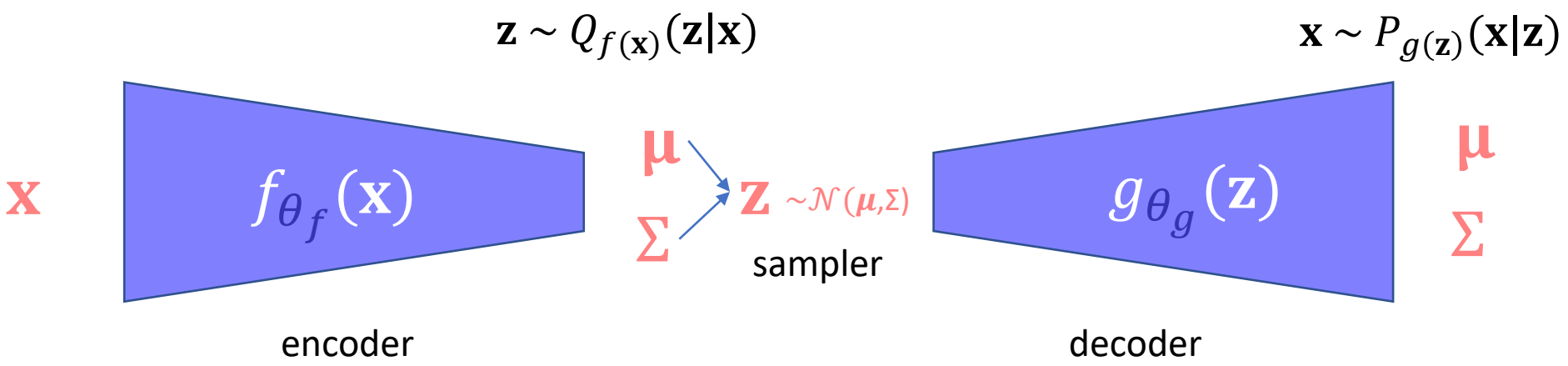
We know the KL of two Gaussians!

Step 5: Further approximations

- Monte Carlo Estimate of the expectation (with *s* samples)
  - $\max_{\theta_f,\theta_g} \sum_{i=1}^{n} \left(\mathbb{E}_{\mathcal{N}(\mathbf{z};\{\boldsymbol{\mu},\Sigma\}=f(\mathbf{x}))} \log \mathcal{N}(\mathbf{x}_i; \{\boldsymbol{\mu},\Sigma\} = g(\mathbf{z})) - D_{KL}(\mathcal{N}(\{\boldsymbol{\mu},\Sigma\} = f(\mathbf{x}_i)) \,||\mathcal{N}(\boldsymbol{\mu}=\mathbf{0},\Sigma=\mathbb{I}) \,) \right)$
  - $$\max_{\theta_f,\theta_g} \sum_{i=1}^{n} \left(\frac{1}{s} \sum_{k=1}^{s} \left(\log \mathcal{N}\left(\mathbf{x}_i; \{\boldsymbol{\mu},\Sigma\} = g\left(\widehat{\mathbf{z}_i^k}\right)\right)\right) - D_{KL}\left(\mathcal{N}(\{\boldsymbol{\mu},\Sigma\} = f(\mathbf{x}_i)) \,||\mathcal{N}(\boldsymbol{\mu}=\mathbf{0},\Sigma=\mathbb{I})\right)\right)$$
    $$\left\{\widehat{\mathbf{z}_i^k}\right\}_{k:1..s} \sim \mathcal{N}(\{\boldsymbol{\mu},\Sigma\} = f(\mathbf{x}_i))$$

VAE - Steps

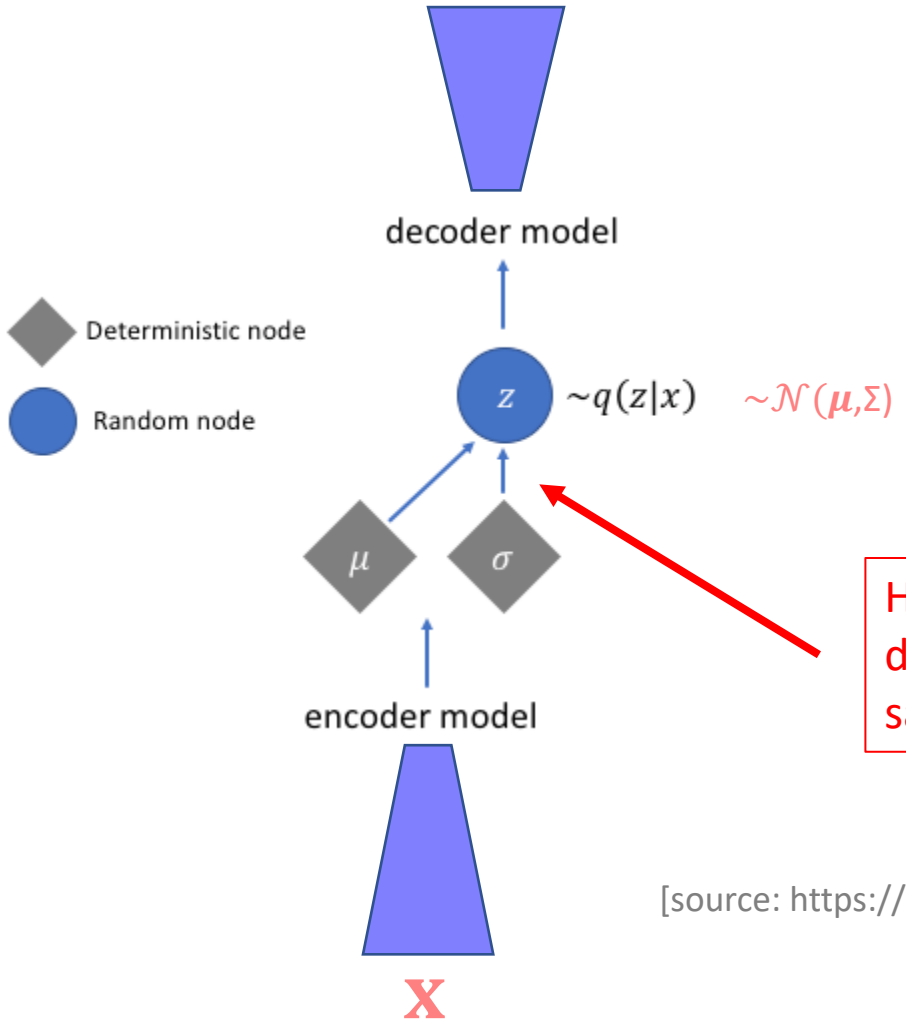
$$\max_{\theta_f,\theta_g} \sum_{i=1}^{n} \left(\frac{1}{s} \sum_{k=1}^{s} \left(\log \mathcal{N}\left(\mathbf{x}_i; \{\boldsymbol{\mu}, \Sigma\} = g\left(\widehat{\mathbf{z}_i^k} \right) \right) \right) - D_{KL}\left(\mathcal{N}(\{\boldsymbol{\mu}, \Sigma\} = f(\mathbf{x}_i)) \,||\mathcal{N}(\boldsymbol{\mu} = \mathbf{0}, \Sigma = \mathbb{I}) \right) \right)$$



But how can we backpropagate through a stochastic node?!

VAE - Steps

Step 6: Reparameterization Trick



How can you take the derivative w.r.t. sampler?

[source: https://www.jeremyjordan.me/variational-autoencoders/]

VAE - Steps

Step 6: Reparameterization Trick

- $$\max_{\theta_f,\theta_g} \sum_{i=1}^{n} \left(\frac{1}{s} \sum_{k=1}^{s} \left(\log \mathcal{N}\left(\mathbf{x}_i; \{\boldsymbol{\mu}, \Sigma\} = g\left(\widehat{\mathbf{z}_i^k} \right) \right) \right) - D_{KL}\left(\mathcal{N}(\{\boldsymbol{\mu}, \Sigma\} = f(\mathbf{x_i})) \,||\mathcal{N}(\boldsymbol{\mu} = \mathbf{0}, \Sigma = \mathbb{I}) \right) \right)$$

$$\widehat{\mathbf{z}_i} \sim \mathcal{N}(\mathbf{z}; \{\boldsymbol{\mu}, \Sigma\} = f(\mathbf{x}_i))$$

Idea

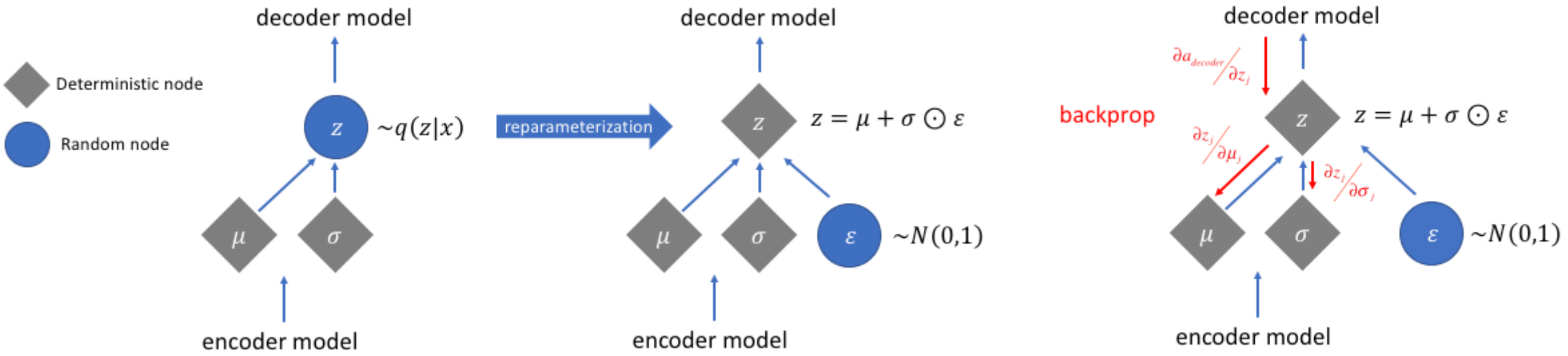
- transfer the stochasticity to a **fixed** distribution (*i.e.*, with no learnable parameters) $Q_{\boldsymbol{\omega}}(\mathbf{z}) \rightarrow Q(\boldsymbol{\varepsilon})$
- Transform samples of the fixed distribution $Q(\boldsymbol{\varepsilon})$ to samples of $Q_{\boldsymbol{\omega}}(\mathbf{z})$ using a parametrized differentiable deterministic function

$$\mathbf{z} = t_{\boldsymbol{\omega}}(\boldsymbol{\varepsilon})$$

Example: $Q(z_i) = N(\mu, \sigma^2)$, then

1. $\varepsilon \sim N(0,1)$
2. $z_i = \sigma\varepsilon + \mu$

Step 6: Reparameterization Trick



[source: https://www.jeremyjordan.me/variational-autoencoders/]